

A cortical-proxy observable for emotionally manipulative media: an instrument, a 400-item corpus, and the field bound it yields

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Abstract

Mean-field opinion models treat media as an external field on a population of coupled states, and that field is almost never measured. A companion paper [8] states what any candidate observable must satisfy for the mechanism to operate: $\alpha \geq h_c(\beta J)/\Delta X$. This paper supplies the missing half, an observable computed from content and the spread it attains on a real corpus. Text is spoken, transcribed, embedded and passed to a released tri-modal encoder that predicts cortical response; two networks are averaged from that prediction, one associated with affective salience and one with deliberative control, and the observable is their signed difference. The ratio form is undefined for 17.25% of items and is replaced by that difference. On 400 items across four categories, matched on length, the categories separate at $\eta^2 = 0.1068$, $F = 15.779$, $p = 1.035 \times 10^{-9}$, against a design powered to detect $\eta^2 = 0.0268$. The separation is carried by fear-activating content, $d = 0.9389$ against neutral, while high-outrage content does not separate, $d = 0.0300$, because it raises both networks by nearly equal amounts. Each category is drawn from one source dataset, so the separation is between corpora and cannot be attributed to framing; we state this before the result rather than in limitations. Against the pre-scan manipulative label the observable reaches $\text{AUC} = 0.6274$, and a TF-IDF baseline on the same rows reaches 0.9758, so the observable is offered as a field quantity and not as a classifier. A second scanning session gives test-retest $\text{ICC} = 0.8725$ and replicates the separation at $\eta^2 = 0.0888$, while 12.8% of individual items reverse

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direction, a count that additive noise at the measured session error predicts, so the instrument supports category-level statements and no per-item verdict. Calibrating the coupling does not identify it: the fit performs worse than the sample mean out of sample and scales with an unmeasured βJ , so no value is quoted. What the measurement does yield is the spread, $\Delta X = 0.1241$, which converts the companion paper’s inequality into a concrete requirement, $\alpha \geq 4.2938$ at $\beta J = 2$, on any study proposing this mechanism.

Keywords: sociophysics, opinion dynamics, external field, media measurement, encoding models, reproducibility

1. Introduction

Statistical-physics treatments of opinion formation routinely include a term representing media. In the mean-field Ising formulation and its relatives [2, 3], that term is an external field h acting on every agent, and the modelling result is that beyond a critical magnitude $h_c(\beta J)$ the population is driven from one consensus to another. The literature on this mechanism is large [4, 5, 6, 7] and the qualitative conclusion is robust within the models.

The field itself is almost never measured. It is assigned, swept, or fitted to an outcome that the model was built to reproduce. A companion paper [8] makes the consequence precise: writing $h = \alpha X$ for a content observable X with attainable range ΔX , no media-driven transition across the spinodal of the all-to-all mean-field model is possible unless

$$\alpha \geq \frac{h_c(\beta J)}{\Delta X}, \quad (1)$$

and this can be evaluated before any data are collected. What it cannot supply is ΔX , which is an empirical property of a real observable on real content.

This paper supplies one. We build an instrument that computes a candidate observable from content, apply it to a corpus assembled for the purpose, and report the spread it attains together with what the measurement does and does not support. Three commitments run through the work and are stated here so the reader can hold us to them: the physics stays in front and the encoder is treated as a thermometer rather than a subject; every number is written by a named script into an artifact and read from there rather than transcribed; and the design’s confound is stated before each result it touches, not deferred.

2. The observable

2.1. Definition

Let A_{aff} be the mean predicted activation over a union of regions associated with affective salience, and A_{del} the corresponding mean over regions associated with deliberative control. The proposal for this work specified a ratio,

$$\text{NAA}_{\text{ratio}} = \frac{A_{\text{aff}}}{A_{\text{del}} + \delta}, \quad (2)$$

which is a magnitude asymmetry only when both means sit at or above baseline. The encoder’s outputs are standardised and unconstrained in sign, and a negative denominator makes the ratio change sign for reasons that have nothing to do with the content. Across the corpus reported below the ratio is undefined for 69 of 400 items, 17.25%. This is not an edge case to be dropped, so the analysis is carried by the signed difference

$$X \equiv \text{NAA}_{\text{signed}} = A_{\text{aff}} - A_{\text{del}}, \quad (3)$$

which is defined everywhere. Positive means affective activation is predicted to lead, negative means deliberative activation leads. The reformulation changes what the index means: it reports which system leads, not by what proportion.

2.2. What the observable is not

The observable is a *cortical proxy*. The released checkpoint predicts activity on the cortical surface only, so no subcortical structure is measured and none is named. No participant was scanned at any point in this work: every value is a prediction produced by an encoder from content, and the phrase “predicted activation” is used throughout in place of any claim about measured brain activity.

3. Instrument and corpus

3.1. The cascade

Each item is converted to speech, transcribed with word-level timings, embedded as text and as audio, and passed to TRIBE v2, a released tri-modal encoder that predicts fMRI response over 20,484 fsaverage5 vertices [1]. The checkpoint is loaded in the average-subject configuration its own

loader forces. Regions of interest are resolved once into vertex index sets and cached, so every item is reduced over identical regions: 1030 vertices for the affective union and 851 for the deliberative union.

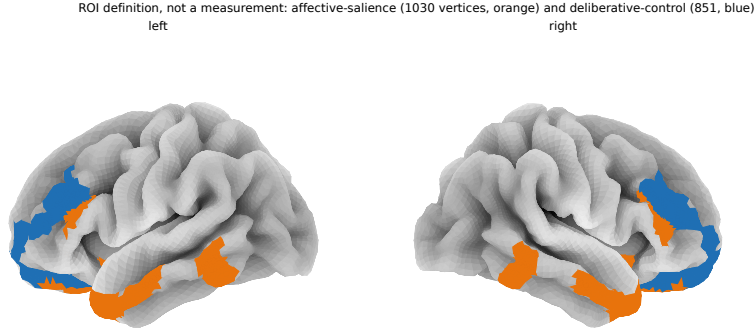


Figure 1: The two regions the observable is reduced over, on the cortical surface. The affective-salience union covers 1030 vertices and the deliberative-control union 851. Both are resolved once into vertex index sets and cached, so every item is averaged over identical regions.

3.2. Corpus design

The corpus is 400 items in four categories of 100, matched on length. Mean word counts are 167.4, 163.5, 163.7 and 162.2 for fear-activating, high-outrage, neutral and reward-hook items respectively, with standard deviations between 10.2 and 12.8, so length is not a confound.

Category	Source	n
Fear activating	ISOT-fake	100
High outrage	SemEval-2019 Task 4	100
Reward hook	Webis-Clickbait-17	100
Neutral informational	PubMed, ISOT-true	50 + 50

Table 1: Corpus composition. Each category is drawn from exactly one source dataset, which is the study’s dominant limitation and is stated before every result it affects. Sources: ISOT [10], SemEval-2019 Task 4 [11], Webis-Clickbait-17 [12], PubMed [13].

The corpus is not source-balanced, and this is the central caveat of the paper. Category and provenance coincide by construction, so any

separation the instrument reports is a separation between corpora and cannot be attributed to framing. Section 4 states this alongside the result rather than after it, and Section 6 identifies which conclusions survive it.

Items were scanned interleaved rather than one category at a time. This turns out to matter, for the reason given in Section 4.4.

4. Results

All values below are read from `data/final/rq_answers.json`, written by `scripts/analyze_corpus.py` from the scan output, which was itself verified row-for-row against the source corpus on text, category and identifier.

4.1. Separation across categories

The four categories differ at

$$F = 15.779, \quad p = 1.035 \times 10^{-9}, \quad \eta^2 = 0.1068, \quad (4)$$

on $n = 400$. The design was fixed before the scan and can detect $\eta^2 = 0.0268$ at $\alpha = 0.05$ and power 0.80, so the observed effect is roughly four times the smallest the study could have found.

Read this with Table 1 in view. The categories are corpora. The measurement establishes that the instrument assigns systematically different values to four groups of documents; it does not establish that framing is what it responds to.

Category vs neutral	t	p	Cohen's d
Fear activating	+6.639	3.283×10^{-10}	+0.939
Reward hook	+2.254	2.539×10^{-2}	+0.319
High outrage	+0.212	8.320×10^{-1}	+0.030

Table 2: Welch comparisons against neutral informational content. The separation is carried by fear-activating items. High-outrage items do not separate, and that is the category the study's hypothesis expected to separate most.

Two features of Table 2 deserve emphasis. First, the effect is one category: remove fear-activating items and little remains. Second, high outrage does not separate on the observable despite being drawn from its own distinct source. Section 4.2 shows why, and the answer is not that outrage content leaves the encoder unmoved.

Every category mean is negative, so deliberative activation is predicted to lead throughout. The instrument finds no affective dominance anywhere in this corpus.

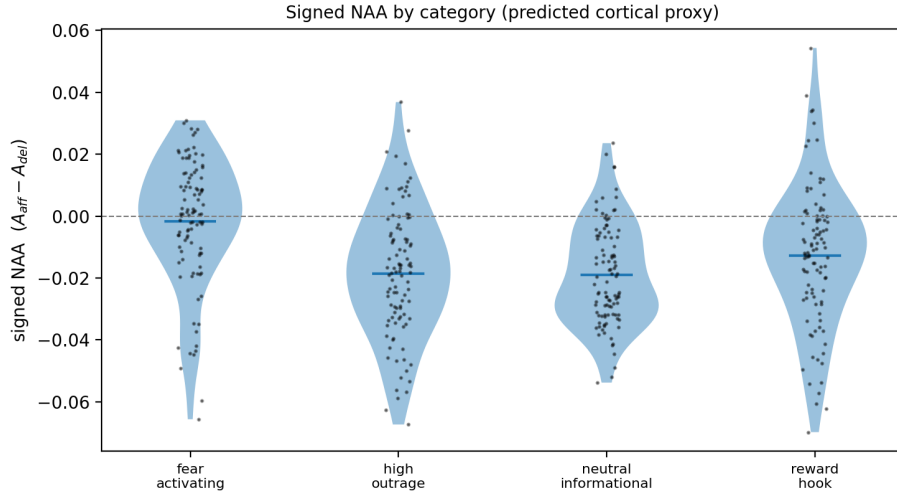


Figure 2: Distribution of the signed observable within each category, all 400 items. Every category mean is negative, so deliberative activation is predicted to lead throughout, and the separation is a shift between overlapping distributions rather than a clean split.

4.2. The two components, taken separately

The observable is a difference, and a difference cannot tell a large symmetric response from no response. Both components are recorded per item, so each can be compared against neutral content on its own.

High outrage raises A_{aff} by $d = +0.507$ and A_{del} by $d = +0.491$, both at $p < 10^{-3}$, and the difference between them by $d = +0.030$. The category that fails to separate on the observable is the one that moves both networks together. Fear-activating content raises A_{aff} by $d = +0.613$ and leaves A_{del} flat, so its whole effect on the observable is one-sided. A one-way ANOVA on each component alone gives $F = 7.451$ for A_{aff} and $F = 6.399$ for A_{del} , both $p < 10^{-3}$.

This weakens an argument we would otherwise make. If outrage had moved neither component, its failure to separate would count against the instrument simply detecting provenance. It moves both, so every category drawn from a distinct source does move away from the baseline on at least

Category vs neutral	Component	t	p	Cohen's d
Fear activating	A_{aff}	+4.333	2.365×10^{-5}	+0.613
Fear activating	A_{del}	-0.696	4.875×10^{-1}	-0.098
High outrage	A_{aff}	+3.584	4.263×10^{-4}	+0.507
High outrage	A_{del}	+3.468	6.521×10^{-4}	+0.491
Reward hook	A_{aff}	+1.842	6.691×10^{-2}	+0.261
Reward hook	A_{del}	-0.063	9.501×10^{-1}	-0.009

Table 3: Welch comparisons of each component against neutral informational content. High outrage moves both components by nearly the same amount; fear-activating content moves one.

one component, which is what source detection predicts. The components do not resolve the confound in either direction.

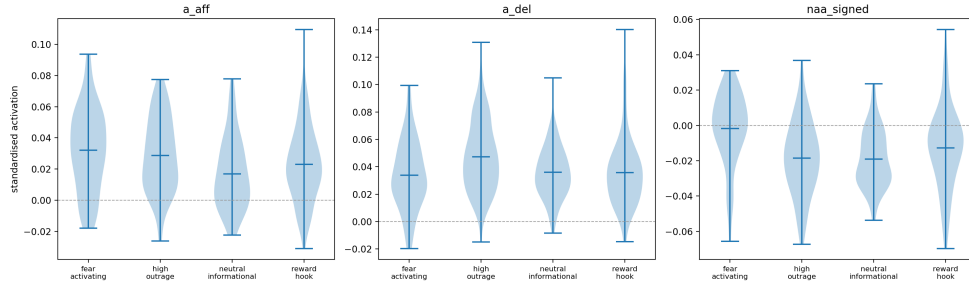


Figure 3: A_{aff} , A_{del} and their signed difference by category over all 400 items. High outrage is elevated on both components and flat on the difference.

4.3. Discrimination against the pre-scan label

Against the **manipulative** label carried by each item before it was scanned, the observable discriminates at $\text{AUC} = 0.6274$, above the smallest detectable 0.5916. AUC is the headline because it is threshold-free. An F_1 of 0.6680 is obtainable, but its threshold was chosen on the same data it is evaluated on, and it is labelled as fitted in sample wherever it appears.

An AUC has no meaning without a baseline, so two were run on the same rows and label by `scripts/baselines.py`: TF-IDF over unigrams and bi-grams with logistic regression, scored out of fold under five-fold stratified

cross-validation, and the VADER [14] compound sentiment score used directly with nothing fitted. Intervals are 2,000-draw bootstrap percentiles.

Model	Split	n	AUC	95% CI
Observable X	nothing fitted	400	0.6274	[0.5702, 0.6867]
VADER compound	nothing fitted	400	0.5392	[0.4757, 0.6035]
TF-IDF + logistic	5-fold stratified	400	0.9758	[0.9601, 0.9879]
Observable X	ISOT only	150	0.7290	[0.6431, 0.8132]
VADER compound	ISOT only	150	0.4019	[0.3098, 0.4943]
TF-IDF + logistic	ISOT only, 5-fold	150	0.9724	[0.9483, 0.9901]

Table 4: Baselines against the same label. The lower block keeps only the 150 items from the ISOT collection, the one origin corpus contributing both classes.

The lexical baseline wins by a wide margin. Sentiment does not clear chance. The observable is not a better detector of manipulative content than counting words, and no such claim is made. Leave-one-source-out, the natural correction for the confound, is unavailable because every source contributes exactly one class, so a held-out source leaves a single-class test fold. The ISOT-only block is the substitute, and TF-IDF holds at 0.9724 there. Both TF-IDF figures nonetheless sit above 0.95, where near-ceiling lexical performance on a small split points to residual stylistic markers, and a probe on the same features recovers the source dataset at 66.25% against 25% chance. Much of what the lexical model separates is provenance. The observable is carried forward as the field quantity Eq. (1) consumes, not as a competing classifier.

4.4. Session reliability

A second GPU session rescanned all 400 items with identical text, identical code and identical region definitions. That pair is a test-retest measurement, and without it every effect above would rest on a single session with no statement of how much of an item’s score is the item and how much is the session.

Three consequences follow, and they cut in different directions.

The separation replicates. On the second session the four-category ANOVA gives $\eta^2 = 0.0888$, $F = 12.861$, $p = 4.918 \times 10^{-8}$, against $\eta^2 = 0.1068$, $F = 15.779$ for the first session over the same items. Both clear the

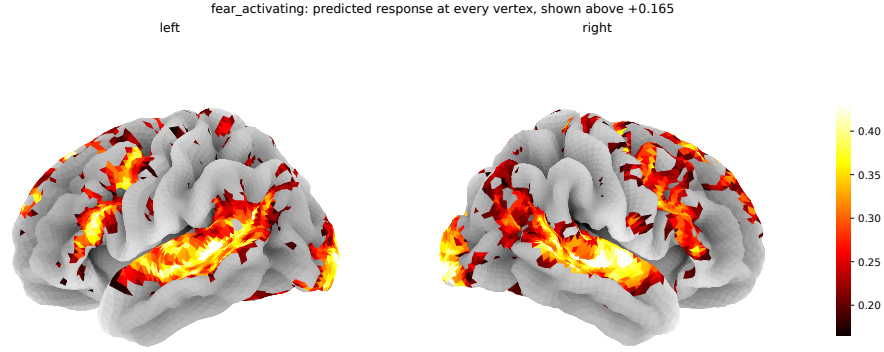


Figure 4: Predicted response for one fear-activating item at every one of the 20,484 cortical vertices, on the pial surface. Vertices below the map’s 70th percentile are left unpainted so the anatomy remains readable, and the threshold is stated in the figure. This is the encoder’s own output, not a smoothed or interpolated field.

	r	ICC(2,1)	$\text{sd}(\Delta)$	$\text{sd}(\Delta)/\text{sd}_{\text{between}}$
$\text{NAA}_{\text{signed}}$	0.8763	0.8725	0.01029	0.484
A_{aff}	0.8759	0.8713	0.01213	0.489
A_{del}	0.8576	0.8550	0.01271	0.521

Table 5: Agreement between two scanning sessions over all 400 items. ICC is reported beside r because a session that shifted every score by a constant would still correlate perfectly.

detectable floor. The result is not an artifact of one session. The second session’s effect is roughly a fifth smaller; both sessions carry the same error, so the difference is within what sampling variation between two noisy runs produces.

Per-item direction is not reliable. Of 400 paired items, 51, or 12.8%, reverse sign between sessions. A statement that a particular item is affective-leading is therefore not supportable at this precision, and no per-item verdict appears anywhere in this paper or in the public interface described in Section 8. Group means over 100 items average this scatter down by an order of magnitude, which is why the category-level result survives.

The reversal count is the measured noise, not an extra defect. The paired differences give a per-session error $\sigma = \text{sd}(\Delta)/\sqrt{2} = 0.00728$, against a between-item standard deviation of 0.01913 once that error is removed. Under additive error at that σ , with each item’s value shrunk toward the grand mean so that no item’s own noise predicts its own reversal, 20,000 simulations give 55.3 expected reversals, 95% interval [44, 67]. The observed 51 sits inside it. Reversing items sit near zero, mean $|X| = 0.00320$ against 0.02159 ($p = 1.4 \times 10^{-25}$), and spread evenly across categories ($\chi^2 = 2.764$, $p = 0.429$). Per-item direction is therefore out of reach at this σ , rather than awaiting a fix.

Scan order does not carry the separation. Because the corpus took more than one session, position in the scan is a candidate explanation for the category effect. It is not one. Items were scanned interleaved, so the four categories are balanced across thirds of the scan order, $\chi^2 = 3.139$, $p = 0.7913$. The drift is real, $\eta^2 = 0.0167$ at $p = 0.0353$, but it is orthogonal to category by construction: removing scan-third means before the category ANOVA moves the effect only from $\eta^2 = 0.1068$ to 0.1048, $p = 1.588 \times 10^{-9}$. Had the corpus been scanned one category at a time, this drift would have been indistinguishable from the finding.

Neither session is corrected toward the other and the two are not averaged. An average of two sessions would carry a precision that neither run has.

5. The physics

5.1. Calibration does not identify the coupling

With the observable in hand, the natural step is to estimate α in $h = \alpha X$ by fitting the model’s predicted outcome against the corpus. On partial runs

the bootstrap interval for $\hat{\alpha}$ comfortably included zero. On the completed corpus it no longer does. That is not a result, for two reasons.

The fit performs worse than the sample mean on held-out data, so it has no out-of-sample predictive value: a negative held-out R^2 means the relationship it encodes does not generalise beyond the points used to estimate it. And the estimate scales inversely with βJ , which nothing in this study measures, so any single value reported would be a statement about a chosen βJ rather than about media.

The coupling is therefore *unidentified*, which is a weaker claim than either a detection or a null. **No value of $\hat{\alpha}$ is quoted in this paper**, in text, table or caption.

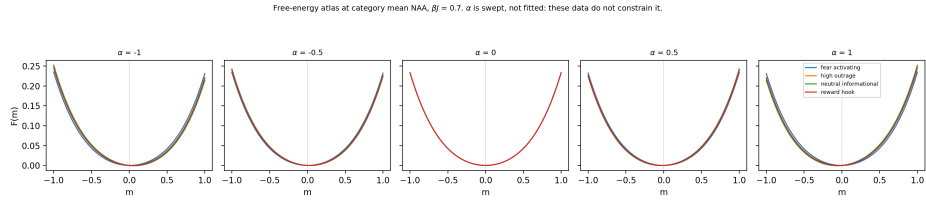


Figure 5: Free energy swept across a range of couplings, evaluated at each category’s measured mean. The sweep is not a fit: no value of α is estimated from the data anywhere in this paper, and the panels show how the physics would behave for each coupling rather than which coupling holds.

5.2. What the spread does yield

The measurement does deliver the quantity Eq. (1) consumes. Across the corpus the observable attains

$$\Delta X = 0.1241, \quad (5)$$

and substituting into the companion paper’s inequality converts a general statement into a concrete requirement.

This is a necessary condition and not a sufficient one. Clearing it does not establish that media moves populations; falling below it excludes the mechanism within the model.

6. What survives the confound

It is worth separating the claims this study can defend from those it cannot, because the two are easy to conflate.

βJ	$h_c(\beta J)$	α required
1.102	0.02101	0.1693
1.496	0.20541	1.6553
2.000	0.53284	4.2938

Table 6: The field bound evaluated at three reduced couplings, using the measured spread $\Delta X = 0.1241$. Each row states the coupling any proposal of this mechanism would need in order for media to drive a transition through this observable.

Not supported. That the instrument detects manipulation, or does so better than a lexical model. That framing rather than provenance drives the separation. Any value for the coupling. Any statement about an individual item. Any claim about subcortical structures, or about measured rather than predicted activity.

Supported. That the observable attains a measured spread of 0.1241 on real content, which is what the bound consumes and which does not depend on why the spread arises. That the categories separate at an effect four times the detectable floor, and that the separation replicates in an independent session. That high-outrage content, the category the hypothesis favoured most, raises both networks together and so leaves the observable where neutral content puts it. And that the coupling is unidentified on this data, which constrains what a successor may assume.

The confound is removable by design rather than by analysis. Drawing all four categories from a single source, so that provenance is held constant while framing varies, or crossing sources with categories so that the two are statistically separable, would convert the present result into an interpretable one. Neither requires new instrumentation.

7. Limitations

1. Category is confounded with source dataset. The dominant limitation, stated in Section 4 as well as here.
2. Single-session scores are noisy: $\text{ICC} = 0.8725$, with session-to-session standard deviation 48.4% of the between-item standard deviation. Category-level conclusions survive this; per-item ones do not.
3. Predicted activation is not measured activation. Whether the encoder tracks real cortex is unresolved and contested; a published audit reports

the released average-subject checkpoint anti-correlated with cortex [15], and a parcel-level replication on public data [9] does not find that sign but is not yet decisive.

4. Cortical proxy only. The checkpoint cannot speak to subcortical structure.
5. Average-subject configuration, forced by the loader.
6. Out-of-distribution stimuli: the encoder was trained on naturalistic audiovisual material and is given synthesised speech from written text.
7. The corpus is English-language from five specific sources, none of them African. A search for openly licensed African news data carrying usable labels returned nothing suitable, so the spread is a property of American and European publishing and the bound inherits that limitation.

8. Conclusion

Media enters mean-field opinion models as a field that is almost never measured. This paper measured a candidate: an observable computed from content through a released cortical encoder, applied to 400 items, with its spread, its reliability and its failure modes reported rather than assumed. The instrument separates four corpora at an effect it was powered to find, the separation replicates across sessions, and it is confounded with provenance in a way no analysis of this corpus can undo. The coupling it was built to calibrate is unidentified. What survives is the spread, and through the companion paper’s inequality a requirement, $\alpha \geq 4.2938$ at $\beta J = 2$, that any future proposal of this mechanism must meet within all-to-all mean-field theory.

Code and data availability

The pipeline, corpus builder, analysis and figure scripts, and the JSON and CSV artifacts every number above is read from, are public at <https://github.com/brn-mwai/monarch>. The instrument runs at <https://monarch-4iy.pages.dev>, carrying the full corpus with each item’s text, its pre-scan labels and its measured values, and reporting no per-item verdict for the reason given in Section 4.4. Figures showing predicted responses are generated by `scripts/render_brain_figures.py` from the saved per-vertex maps, each stating its threshold, rather than captured from the interface.

CRedit authorship contribution statement

Brian Mwai: Conceptualization, Methodology, Software, Validation, Formal analysis, Investigation, Data curation, Writing – original draft, Writing – review and editing, Visualization. The author is the sole contributor to this work.

Declaration of competing interest

The author declares no competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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